Software Requirements Specification

-

E-Consultation Mobile App

-

Submitted to

Ministry of Innovation and Technology

Document Version 2

Aug 12, 2025

Prepared By

360Ground™ ELDIX IT Technology PLC

Table of Contents

1 Introduction 3

1.1 Purpose 3

1.2 Scope 4

1.3 Intended Audience 4

1.4 Definitions, Acronyms, and Abbreviations 5

2 Specific Requirements 6

2.1 Assumption 6

2.2 Functional Requirements (FR) 7

2.2.1 User Registration 7

2.2.2 Document Access and Exploration" 8

2.2.3 Public Consultation & Feedback 11

2.2.4 API Integration and Backend Communication 14

2.3 Non-Functional Requirements (NFR) 16

1 Introduction

This System Requirements Specification (SRS) document serves as the official blueprint for the development of the E-Consultation Mobile App. The project's core objective is to significantly enhance citizen participation and government transparency by offering a streamlined and intuitive mobile experience. The app will serve as the user-facing platform, providing citizens with easy access to draft legal documents and a direct channel to submit their valuable feedback, thereby contributing to the law-making process.

The E-Consultation Mobile App is a strategic initiative to digitize and improve the legislative process. By providing a convenient mobile platform, the project aims to foster a more inclusive and democratic environment where all citizens can easily engage with the legal system. The development of this app aligns with the government's broader objectives of leveraging technology to improve public services and governance.

The platform will provide citizens with an innovative and transparent mechanism to view and comment on draft laws and regulations. It will offer users a user-friendly and secure environment to engage in public consultations, while promoting improved transparency and citizen-centric governance. The app is a vital component of the strategy to deepen public engagement and develop a sustainable and well-regulated law-making ecosystem.

The SRS document covers the complete scope of the platform's functionalities, including user registration, document viewing, and feedback submission. Additionally, it addresses the technical and non-technical requirements needed to ensure the app's success in supporting the government's goals and its stakeholders. The platform will serve as an important tool to foster public participation, transparency, and trust, contributing significantly to the nation's broader governance goals.

1.1 Purpose

The purpose of this document is to provide a detailed description of the functional and non-functional requirements for the **E-Consultation Mobile App**. This app is a key component of the Ministry of Justice's digital transformation strategy, under the broader mGov initiative led by the Ministry of Innovation & Technology.

The primary goal of this application is to foster greater public participation, transparency, and efficiency in the legislative process. It will serve as a direct, user-friendly channel for the government to publish draft laws (such as proclamations, directives, and regulations) and for citizens to provide structured, timely feedback. This document is intended to be the single source of truth for all project stakeholders, ensuring a common understanding of the system to be built.

This document serves several key purposes:

1. **Establish a Clear Understanding**: To provide a clear and detailed understanding of the platform's features, functionalities, and expected outcomes for stakeholders, including project sponsors, developers, system administrators, and end users.
2. **Guide the Development Process**: To serve as the foundational document that guides the software development team through the design, development, testing, and implementation phases, ensuring that the platform meets the business objectives and the technical standards required.

3. **Define System Scope and Requirements**: To outline the scope of the platform, including its functional components such as user registration, document access, and feedback submission.
4. **Facilitate Stakeholder Alignment**: To ensure alignment between all project stakeholders—Ministry of Justice's, MInT, 360Ground, and other partners—by detailing the platform's requirements and clarifying expectations around platform functionalities, performance, security, and compliance.
5. **Ensure Compliance and Quality**: To define the non-functional requirements such as security, scalability, performance, and usability, ensuring the platform adheres to industry standards, regulatory requirements, and best practices, and delivers a secure, reliable, and user-friendly experience.
6. **Basis for Testing**: It provides a basis for creating test cases and scenarios to ensure that the platform functions as intended and meets the quality standards. And it serves as a reference point for validating and verifying the final product, ensuring that the developed platform meets the specified requirements.

1.2 Scope

The E-Consultation Mobile App is designed as a client-side application for the existing web portal. Its scope is limited to the following user-centric functionalities:

- **User Management:** This includes user registration, secure login, and the ability for users to manage their profiles.
- **Document Access:** Users will be able to browse, search, and view draft legal documents.
- **Public Feedback:** Authenticated users will be able to submit general comments on a document or specific comments on individual articles.

The following administrative functionalities are explicitly **out of scope** for the mobile app and will continue to be handled by the existing web portal:

- Publication of legal documents.
- Administrative user management.
- Moderation of comments.
- Reporting and analytics.

All communication between the mobile app and the backend portal will be handled through a secure API, ensuring data integrity and consistency across platforms.

1.3 Intended Audience

This document is intended for a wide range of stakeholders involved in the development, management, and oversight of the E-Consultation Mobile App project, including:

- **Project Managers:** To understand the project scope, plan resources, and track progress.
- **Software Developers (Mobile & Backend):** As a technical blueprint for designing, building, and implementing the system.
- **Quality Assurance (QA) Engineers:** To create test cases and verify that the app meets all specified requirements.
- **Ministry of Justice Officials (Product Owners/Admins):** To confirm that the specified features align with their policy goals and operational needs.
- **Ministry of Innovation & Technology Stakeholders:** To ensure the project aligns with the national mGov strategy and technical standards.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
Citizen	A general, registered user of the app who can view documents and provide feedback.
Draft	A legislative document that is open for public consultation.
mGov	Mobile Government; the use of mobile technologies to deliver government services and information.
NFR	Non-Functional Requirement; defines how a system should operate, rather than what it should do.
Push Notification	A message that pops up on a mobile device to inform users of new or important information from an application, even when the app is not in use**.**
OTP	One-Time Password; a password that is valid for only one login session or transaction.
API	Application Programming Interface; a set of rules and protocols for building and interacting with software applications
Backend Portal	The existing web-based system that handles administrative functions, data management, and content publication for the mobile app**.**
bcrypt	A strong, adaptive password-hashing function used to securely store passwords.
Cross-Site Scripting (XSS)	A type of security vulnerability where malicious scripts are injected into otherwise benign and trusted websites or applications.
SQL Injection	A code injection technique used to attack data-driven applications, in which malicious SQL statements are inserted into an entry field for execution.
PDP	Personal Data Protection Proclamation of Ethiopia.
SRS	System Requirements Specification.
UI	User Interface.
User Story	A short, simple description of a feature told from the perspective of the person who desires the new capability, in the format: "As a <role>, I want <goal>, so that <reason>."

2 Specific Requirements

This section provides the detailed requirements for the E-Consultation Mobile App. It is the core of the SRS, specifying every feature, function, and operational constraint that the system must satisfy. To ensure a complete and clear specification, these requirements are organized into two essential and distinct categories: **Functional Requirements** and **Non-Functional Requirements**.

- **Functional Requirements** define the explicit behaviors and capabilities of the system—in other words, ***what the application must do***. They are presented as a comprehensive set of user stories from the perspective of each user role. This user-centric format ensures every feature is tied to a clear goal and delivers tangible value. Each story is accompanied by precise, testable acceptance criteria that serve as a definitive checklist for development and quality assurance.
- **Non-Functional Requirements (NFRs)** complement the functional specifications by defining the system's quality attributes and operational standards—that is, ***how well the system must perform its functions***. These are the critical requirements that govern the user experience, making the app reliable, secure, and easy to use. This category covers crucial aspects such as performance targets, security protocols, usability standards, and compatibility across different devices.

2.1 Assumption

The requirements outlined below are fully dependent on the existing E-Consultation Portal and its backend technologies (API). For these user stories to be implemented effectively, it is assumed that the existing backend has a fully functional API capable of supporting all the specific functionalities described in the following User Stories section.

It should be clearly noted that if any feature described below is not available via the API, it will be assumed that such functionality will not be implemented in the mobile application, and that the development or modification of the API falls outside the scope and responsibility of the consultant.

2.2 Functional Requirements (FR)

2.2.1 User Registration

ID	Title	User Role	User Story	Acceptance Criteria	Priority
UR-001	**Guest Access to Documents**	General User	As a General User, I want to browse, search, and read all published legal documents	• The user can open the app and view a list of all documents with "Draft" or "Finalized" status	**High**

ID	Title	User Role	User Story	Acceptance Criteria	Priority
		(Unauthenticated)	without needing to create an account, so that I can stay informed easily and barrier-free.	• The user can tap on any document to open and read its full content • The user can use the search and filter functionalities • Commenting, or any feedback submission buttons are either hidden or prompt the user to "Log in or Register" when tapped	

UR-002	**User Account Registration**	General User	As a General User, I want to create an account using my email or phone number, so that I can participate in consultations by providing feedback on draft laws.	• The registration screen must provide options for signing up with either an **email** address or a **phone number**.
				• The form must require:
				  ○ First name
				  ○ last name
				  ○ email/phone
				  ○ password.
				• A "Confirm Password" field is required and must match the "Password" field.
				• A verification step (OTP sent to phone or verification link to email) is required to activate the account.
				• Upon successful verification, the user is automatically logged in with a 'Citizen' role.
					High

UR-003	**User Login**	Registered user	As a registered user, I want to log in to my account with my credentials, so that I can access features specific to my role.	• The login screen must have fields for email/phone and password • Upon correct login, the user is taken to the main app screen • The login screen must include a "Forgot Password?" link	**High**
UR-004	**Password Reset**	Registered user	As a registered user, I want to reset my password if I forget it, so that I can regain access to my account securely.	• The user can initiate the password reset process from the login screen • The system sends a time-sensitive, single-use reset code to their registered email or phone • The user enters the code in the app to verify their identity • The app then prompts the user to enter and confirm a new password.	**High**
UR-005	**User Logout**	Logged-in user	As a logged-in user, I want to log out of my account, so that I	• A clear "Logout" option is available within a user profile or settings menu	**Medium**

ID	Title	User Role	User Story	Acceptance Criteria	Priority
			can ensure my session is secure.	• Tapping "Logout" immediately terminates the user's session and clears any sensitive local data • The user is returned to the public/guest view of the app.	
UR-09	**View/Edit User Profile**	Registered user	As a registered user, I want to view and edit my own profile information, such as my name, so I can keep my personal details up to date.	• A "My Profile" or "Account" section must be accessible to logged-in users • This screen displays the user's registered full name and email/phone number • The user must be able to edit their full name • The user cannot edit their registered email/phone number as it is their primary identifier • The screen should provide an option to "Change Password"	**Medium**

2.2.2 Document Access and Exploration"

ID	Title	User Role	User Story	Acceptance Criteria	Priority

PLD-001	**View List of Documents**	General User / Citizen	As any user, I want to see a clear, organized list of all legal documents on the app's main screen, so I can quickly find what I'm looking for.	• The app's home screen displays a list of documents
				• Each list item must show:
				  ○ Title
				  ○ Institution
				  ○ Category
				  ▪ Proclamation
				  ▪ Directive
				  ▪ Regulations etc.
				  ○ Status
				  ▪ Open for Consultation
				  ▪ Closed
				• Documents are sorted by the most recently published or updated by default
				• A clear visual indicator (colored tags) distinguishes between statuses
				• The list supports **infinite scrolling/Explore more button.**
					High

ID	Title	User Role	User Story	Acceptance Criteria	Priority

PLD-002	**View Document Details**	General User / Citizen	As any user, I want to tap on a document from the list to view its full details and content, so I can read it thoroughly before commenting.	• Tapping a document opens a new detail screen
				• This screen must display:
				  ○ Title
				  ○ Summary
				  ○ Category
				  ○ Status (Draft, Open for Consultation, Finalized)
				  ○ Full Document Content,
				  ○ Publishing Authority
				  ○ Open for Consultation Date
				  ○ Closed for Consultation Date
				  ○ A "Download Document" button to save the file for offline reading**.**
				• The user can scroll through the entire document content smoothly.
					High

PLD-003	**Search & Filter Documents**	General User / Citizen	As a user, I want to search with keywords and apply filters to the document list, so I can quickly and efficiently find the specific legislation I am interested in.	**Search Functionality:**
				• A search bar must be prominently displayed on the main document list screen.
				• The search must query the document **Title**, and **Publishing Authority** fields for the keyword.
				• The search must be case-insensitive and should start filtering the list dynamically as the user types.
				Filter Functionality:
				• A "Filter" button must be available, opening a panel with filtering options.
				• The filter panel must allow filtering by:
				  ○ **Category/Type** (multi-select list).
				  ○ **Status** (multi-select list).
				  ○ **Publication Date** (pre-set ranges and a custom date picker).
				• The user can apply both a search term and filters simultaneously.
				• A clear "Reset" or "Clear All" option must be present to remove all active search and filter criteria, returning the list to its default state.
				• A visual indicator must show when any search or filter is active.
					High

PLD-004	**Multilingual Support**	General User / Citizen	As a user, I want to experience the	**Interface Language:**	**Low**

ID	Title	User Role	User Story	Acceptance Criteria	Priority
			entire app, from its interface to its document content, in my preferred language (Amharic or English), so that the app is fully accessible to me.	• The app must provide an option in Settings/ Home Screen to switch the entire UI language between Amharic and English • All static text (buttons, menus, labels) must respect the selected language • The user's language preference must be saved for future sessions. **Content Language:** • When viewing a document with multiple language versions, clear toggle buttons ("English" "Amharic") must be displayed. • Tapping a language toggle must instantly switch the displayed document content to that language's version • If a document is available in only one language, no language toggles should be shown.	
PLD-005	**Push Notifications**	Citizen	As a user, I want to receive push notifications for important events related to new or updated laws, so I can stay informed in real-time.	**Notification Triggers:** • A push notification must be automatically sent to all users when a new draft law being posted on the backend portal. • A push notification must be sent to all users 24 hours before a public consultation deadline. • A push notification must be sent to all users when the status of a draft law changes. **Notification Content:** • The notification title must be clear and descriptive. • The notification body must include the name of the relevant document and a brief, one-sentence summary of the event. **User Interaction:** • Tapping on a push notification must open the app and take the user directly to the details page of the relevant document. **User Control:** • The app must provide a setting for users to enable or disable push notifications. This includes a general toggle for all notifications.	**High**
PLD-003	**Download Document**	General User / Citizen	As any user, I want to download a document as a PDF,	• On the document details screen, a "Download" button should clearly visible • Tapping it initiates a PDF download	**Medium**

ID	Title	User Role	User Story	Acceptance Criteria	Priority
			so I can read it offline or share it.	• The mobile OS handles saving the file • A confirmation message is shown.	

2.2.3 Public Consultation & Feedback

ID	Title	User Role	User Story	Acceptance Criteria	Priority
PCF-001	**View Comments**	General User / Citizen	As any user, I want to read the comments and replies on a draft document, so I can understand public opinion and different perspectives.	• On the detail screen of a document "Open for Consultation", a "Comments" section is visible. • **For General Comments:** The section displays a list of all comments made on the document as a whole, sorted by newest first by default. Each comment shows the commenter's name (or "Anonymous Citizen"), the comment text, and the date it was posted. Any replies to a comment are displayed directly beneath it, visually indented. • **For Per-Article Comments:** The section displays a list of comments specifically tagged to a particular article. Each comment will clearly indicate the article it refers to It will also show the commenter's name (or "Anonymous Citizen"), the comment text, and the date it was posted. Replies are nested under the parent comment. • The list of comments for both general and per-article comments supports lazy loading (infinite scroll).	**High**
PCF-002	**Submit a Comment**	Citizen	As a registered Citizen, I want to submit my feedback either as a general comment on the entire draft law or as a	**General Commenting:** • A logged-in 'Citizen' viewing a document "Open for Consultation" must see an **"Add**	**High**

ID	Title	User Role	User Story	Acceptance Criteria	Priority
			specific comment tied to a particular article, so my opinion can be considered with appropriate context.	**General Comment"** button or a similar call to action. • Tapping this button opens a simple text input area (max 2000 characters) for the user to provide feedback on the document as a whole. • If anonymous commenting is enabled, a "Post anonymously" checkbox is visible. • Upon submission, the comment is posted publicly and appears in the main comments list. **Per-Article Commenting:** • When a 'Citizen' is viewing the full document content, each article or section should be selectable or have a dedicated "Comment" button next to it. • Tapping this button opens a simple text input area (max 2000 characters). The comment is automatically associated with the specific article. • This comment will be displayed within the main comments feed, but with a clear visual tag or indicator that it is linked to a specific article Upon submission of either type of comment, a confirmation message appears, - "Your comment has been posted."	
PCF-003	**Reply to a Parent Comment**	Citizen	As a registered Citizen, I want to reply directly to another person's main comment to engage in discussion.	• Each top-level, or "parent," comment card in the comments list must have a clearly visible "Reply" button. • Tapping the "Reply" button on a parent comment will open a comment	**Medium**

ID	Title	User Role	User Story	Acceptance Criteria	Priority
				submission interface, such as a text input field or a pop-up, specifically for writing a reply. • This reply interface will be visually linked or "contextually aware" of the parent comment it is responding to. • The maximum character limit for a reply should be 2000 characters, consistent with the main comment submission. • Upon submitting the reply, it is posted publicly and appears immediately, nested or visually indented directly beneath the parent comment. • A confirmation message ("Your reply has been posted.") should be displayed. • This feature is only available to logged-in 'Citizen' users.	
PCF-004	**Vote on Comments**	Citizen	As a registered Citizen, I want to upvote or "like" comments that I agree with, so I can quickly express my support for an opinion without having to write a full comment myself.	**Narration:** • While reading through the comments, a user sees an opinion they strongly agree with. • They simply tap the "thumbs-up" icon next to the comment. • The icon fills with color, and the number next to it increases by one, giving them a quick and easy way to participate. **Key Considerations:** • Next to each comment and reply, there must be a clearly visible upvote/like button (a "thumbs-up" icon) and a counter displaying the total number of votes.	**Medium**

ID	Title	User Role	User Story	Acceptance Criteria	Priority
				• This feature is only enabled for logged-in 'Citizen' users. • A user can tap the vote button on any comment (except their own) to cast their vote. • When a user casts a vote, the button's visual state must change (icon becomes solid or changes color) to provide instant feedback that their vote was registered. • The vote count next to the button must immediately increment by one. • Tapping the button again will "un-vote" or withdraw the vote. • The button reverts to its original state, and the count decrements by one. • The system must prevent a user from voting on their own comments; the button should be disabled or not visible on their own comments.	

2.2.4 API Integration and Backend Communication

ID	Title	User Role	User Story	Acceptance Criteria	Priority
API-001	**API Integration**	System (Mobile App)	As the mobile app, I need to securely communicate with the backend portal to fetch, send, and receive all necessary data, so I can function as the user-facing component of the e-consultation system.	• **Authentication:** The mobile app must use a secure authentication method (OAuth 2.0 or JWT) to interact with the backend API. • **Endpoints:** The app must be able to call the following endpoints on the backend portal: ○ `GET /documents`**:** To fetch the list of all legal documents with their details.	**Medium**

ID	Title	User Role	User Story	Acceptance Criteria	Priority
				○ `GET /documents/{id}`: To retrieve the full content and details of a specific document. ○ `GET /documents/{id}/comments`: To fetch all comments and replies for a specific document. ○ `POST /comments`: To submit a new general or per-article comment from a registered user. ○ `POST /comments/{id}/reply`: To submit a reply to a parent comment. ○ `POST /comments/{id}/vote`: To cast a vote (upvote/like) on a specific comment. ○ `POST /users/register`: To create a new user account (Citizen). ○ `POST /users/login`: To authenticate and log in a registered user. ○ `POST /users/password-reset`: To initiate a password reset request. ○ `GET /users/me`: To retrieve and update the current user's profile information. • **Data Format:** All data sent to and from the API must be in a standardized format, preferably JSON. • **Error Handling:** The app must be able to gracefully handle API errors and provide clear, user-friendly messages. • **Security:** All API communication must be encrypted using HTTPS (TLS 1.2 or higher), and sensitive data should not be stored in an insecure manner on the device.	

2.3 Non-Functional Requirements (NFR)

Category	Requirement
1. Performance	• **Fast Response Time:** User actions must provide feedback in under 2 seconds. The initial document list should load in under 3 seconds on a 4G connection.
	• **Efficient Resource Usage:** The app must minimize CPU, memory, and battery consumption. The installed app size should target less than 50MB.
	• **Low-Bandwidth Operation:** The app must be fully functional on slow (3G) network connections by optimizing data and image loading.
2. Security	• **Secure Data Transmission:** All communication between the app and server must be encrypted using HTTPS (TLS 1.2 or higher).
	• **Secure Authentication:** User passwords must be securely salted and hashed on the server (e.g., using bcrypt). Passwords must never be stored in plaintext.
	• **Data Protection:** Sensitive data stored on the device (e.g., authentication tokens) must be encrypted using OS-level security (iOS Keychain, Android Keystore).
	• **Vulnerability Protection:** The system must be protected against common web and mobile vulnerabilities, including SQL Injection and Cross-Site Scripting (XSS).
3. Usability	• **Intuitive Navigation:** The app's layout and workflow must be self-explanatory and follow standard mobile OS design patterns. A new user should not require a tutorial.
	• **Clarity of Language:** All text within the app (UI labels, instructions, error messages) must be clear, concise, and easy to understand in all supported languages.
4. Reliability & Availability	• **High Availability:** The backend services must maintain a minimum uptime of 99.5%.
	• **Graceful Error Handling:** If an error occurs (e.g., loss of network), the app must inform the user with a clear message and not crash.
5. Compatibility	• **Operating System Support:** The app must be fully compatible with the two most recent major versions of iOS and with Android 8.0 (Oreo) and newer.
	• **Device Support:** The app's UI must adapt correctly to various screen sizes and resolutions, including both standard phones and tablets.
6. Scalability	• **User Load:** The backend system must be able to handle at least 100,000 concurrent users without performance degradation.
	• **Data Volume:** The system architecture must be able to efficiently manage and serve a growing volume of data.

Category	Requirement
7. Maintainability	• **Modular Architecture:** The code must be well-structured and modular to allow for easy updates, feature additions, and bug fixes without major re-writes.
	• **Code Documentation:** The source code should be clearly commented and documented to facilitate knowledge transfer and future development.
8. Legal & Compliance	• **Data Privacy Law Compliance:** The app must fully comply with the Ethiopian Personal Data Protection Proclamation (PDP) and other applicable regulations.
	• **Terms of Service & Privacy Policy:** The app must provide users with easy access to its Terms of Service and a clear Privacy Policy, as defined in functional requirement GAF-001.